Fundamental Algorithmic Techniques XI

April 16, 2026

Outline

Linear Programming: Examples

Primal, Dual & Optimisation

Simplex Algorithm

Linear Programming: Examples

Formulation of Optimisation Problems with constraints:

Example 1: Bakery Production

- **Decision Variables:**
  $x =$ croissants, $y =$ muffins
- **Objective:**
  max Profit $= 3x + 4y$
- **Constraints:**

$$\begin{aligned} 2x + 3y &\leq 100 \quad \text{(Flour)} \\ x + 2y &\leq 50 \quad \text{(Sugar)} \\ 3x + 2y &\leq 80 \quad \text{(Labor)} \\ x, y &\geq 0 \end{aligned}$$

Example 2: Manufacturing

- **Decision Variables:**
  $b =$ basic product, $p =$ premium product
- **Objective:**
  max $P = 30b + 40p$
- **Constraints:**

$$\begin{aligned} 20b + 30p &\leq 3900 \quad \text{(Res 1)} \\ 15b + 30p &\leq 3300 \quad \text{(Res 2)} \\ b, p &\geq 0 \end{aligned}$$

Transportation Problem: Example

Cost Matrix, Supply & Demand

	D_1	D_2	D_3	D_4	Supply
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	**34**

LP Formulation:

- **Decision Variables:** x_{ij} = units shipped from S_i to D_j $\quad (i = 1, 2, 3;\ j = 1, 2, 3, 4)$
- **Objective:**

$$\min Z = 19x_{11} + 30x_{12} + 50x_{13} + 10x_{14} + 70x_{21} + 30x_{22} + 40x_{23} + 60x_{24} + 40x_{31} + 8x_{32} + 70x_{33} + 20x_{34}$$

Supply Constraints:

$$x_{11} + x_{12} + x_{13} + x_{14} = 7$$
$$x_{21} + x_{22} + x_{23} + x_{24} = 9$$
$$x_{31} + x_{32} + x_{33} + x_{34} = 18$$

Demand Constraints:

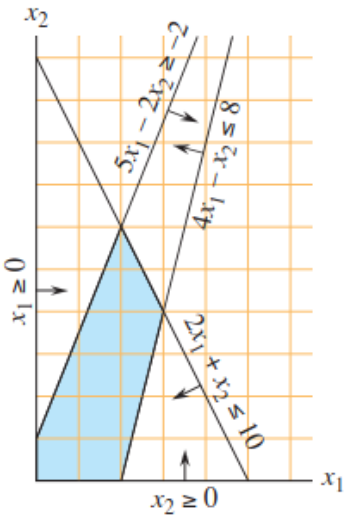
$$x_{11} + x_{21} + x_{31} = 5$$
$$x_{12} + x_{22} + x_{32} = 8$$
$$x_{13} + x_{23} + x_{33} = 7$$
$$x_{14} + x_{24} + x_{34} = 14$$

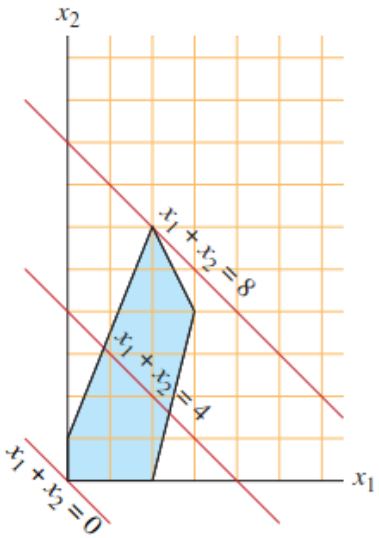
Non-negativity: $x_{ij} \geq 0 \quad \forall i, j$

Example: A two-variable linear program

Formulation:

$$
\begin{aligned}
\text{maximize} \quad & 2x_1 + 3x_2 \\
\text{subject to} \quad & -x_1 + x_2 \leq 8 \\
& 2x_1 + x_2 \leq 10 \\
& 5x_1 - 2x_2 \geq -2 \\
& x_1, x_2 \geq 0
\end{aligned}
$$





Key Ideas:

- Each constraint $\rightarrow$ a half-plane
- Feasible region = intersection (convex polygon)
- Optimal solution at a **vertex**

General Linear Programming Formulation

Core Idea: Optimize a linear objective subject to linear constraints. *(Strict inequalities not allowed in LP.)*

Standard Maximization Form:

$$\begin{aligned} \max \quad & Z = \sum_{j=1}^{n} c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^{n} a_{ij} x_j \leq b_i \quad \forall i = 1, \ldots, m \\ & x_j \geq 0 \quad \forall j = 1, \ldots, n \end{aligned}$$

Compact Matrix Notation:

$$\begin{aligned} \max \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{c}, \mathbf{x} \in \mathbb{R}^n$.

- **Objective:** $\mathbf{c}^\top \mathbf{x}$ — linear combo of decision variables
- **Constraints:** m linear inequalities + n non-negativity bounds
- **Feasible Region:** Intersection of half-spaces $\Rightarrow$ convex polyhedron

Introduction to Primal / Dual Space

Primal (Max)

$$\begin{aligned} \max \quad & 3x_1 + x_2 + 4x_3 \\ \text{s.t.} \quad & x_1 + x_2 + 3x_3 \leq 30 \\ & 2x_1 + 2x_2 + 5x_3 \leq 24 \\ & 4x_1 + x_2 + 2x_3 \leq 36 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Dual (Min)

$$\begin{aligned} \min \quad & 30y_1 + 24y_2 + 36y_3 \\ \text{s.t.} \quad & y_1 + 2y_2 + 4y_3 \geq 3 \\ & y_1 + 2y_2 + y_3 \geq 1 \\ & 3y_1 + 5y_2 + 2y_3 \geq 4 \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$$

Primal–Dual Relationships:

- **Constraint $\leftrightarrow$ Variable:** Each primal constraint corresponds to a **dual variable** (y_i), and vice versa.
- **Coefficient mapping:** The primal objective coefficients become RHS of dual constraints; primal RHS becomes dual objective coefficients.
- **Weak duality:** For any feasible x and y: $c^\top x \leq b^\top y$.
- **Strong duality:** If both are feasible, optimal values are equal: $c^\top x^* = b^\top y^*$.
- **Complementary slackness:** At optimum, $x_j > 0 \Rightarrow j$-th dual constraint is tight, and $y_i > 0 \Rightarrow i$-th primal constraint is tight.

Fundamental Theorem of Linear Programming

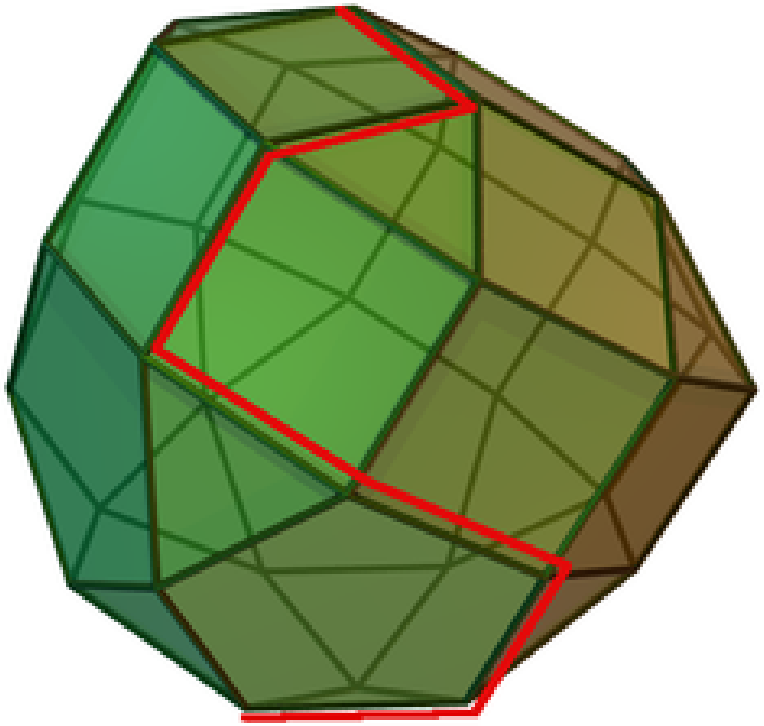
Theorem Any linear program in standard form satisfies **exactly one** of the following:

1. **Optimal:** Has an optimal solution with finite objective value.
2. **Infeasible:** No point satisfies all constraints (feasible region $= \emptyset$).
3. **Unbounded:** Feasible region is non-empty, but objective can be made arbitrarily large (for max) or small (for min).

Implications:

- Guarantees that LP solvers will terminate with a definitive answer
- Underpins the correctness of the simplex method and interior-point algorithms
- Justifies checking feasibility and boundedness before seeking optimality

Simplex Algorithm: Geometric Intuition



Source: Wikipedia / Simplex method in 3D

Core Idea:

- Exploit problem geometry in primal & dual space
- Navigate along vertices of the feasible polyhedron
- Each pivot moves to a better adjacent vertex

How Simplex Works:

- Starts at a feasible vertex
- Pivots to adjacent vertex with better objective
- Stops at optimum (no improving neighbor)

Simplex Algorithm and Tableau

Primal (Max):

$$\begin{aligned} \max \quad & Z = c_1 x_1 + c_2 x_2 \\ \text{s.t.} \quad & a_{11} x_1 + a_{12} x_2 \leq b_1 \\ & a_{21} x_1 + a_{22} x_2 \leq b_2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Dual (Min):

$$\begin{aligned} \min \quad & W = b_1 y_1 + b_2 y_2 \\ \text{s.t.} \quad & a_{11} y_1 + a_{21} y_2 \geq c_1 \\ & a_{12} y_1 + a_{22} y_2 \geq c_2 \\ & y_1, y_2 \geq 0 \end{aligned}$$

Intro resource:
`matem.unam.mx/simplex-intro`

Initial Simplex Tableau:

BV	x_1	x_2	s_1	s_2	RHS
s_1	a_{11}	a_{12}	1	0	b_1
s_2	a_{21}	a_{22}	0	1	b_2
Z	$-c_1$	$-c_2$	0	0	0

How Simplex Works:

- Starts at feasible vertex (slacks s_1, s_2)
- Pivots to adjacent vertex with better objective
- **Pivot rule**: Most negative in Z-row + min-ratio test
- **Optimal**: All Z-row coefficients ≥ 0

Simplex Algorithm: One Iteration

Initial Tableau:

BV	x_1	x_2	s_1	s_2	RHS
s_1	a_{11}	a_{12}	1	0	b_1
s_2	a_{21}	a_{22}	0	1	b_2
Z	$-c_1$	$-c_2$	0	0	0

After Pivot (Example):

BV	x_1	x_2	s_1	s_2	RHS
x_1	1	α	β	0	γ
s_2	0	δ	ϵ	1	ζ
Z	0	η	θ	0	Z_1

Coefficients updated via Gauss–Jordan elimination.

Resolution Steps:

1. **Optimality Test:** Are all Z-row coeffs ≥ 0? *Yes* $\rightarrow$ STOP (optimal). *No* $\rightarrow$ continue.
2. **Entering Variable:** Pick column with most negative Z-row entry (e.g., x_1 if $-c_1 < -c_2$).
3. **Leaving Variable (Min-Ratio Test):**
   $$\min \left\{ \frac{b_i}{a_{i,\text{enter}}} \;\middle|\; a_{i,\text{enter}} > 0 \right\}$$
   Row achieving min $\rightarrow$ leaving basic variable.
4. **Pivot:**
   - Normalize pivot row: divide by pivot